

Introduction to Programming with Scientific Applications

- Missing handins, deadline 15 May 2026
- Final project, deadline 31 May 2026
- Course evaluation
- Exam, 20 June 2026
- AOB

Exam

- 5 hours, written exam, with aids, including PC but **without internet**
- Communication with others about the exam is not permitted during the exam
- WISEflow Device Monitor must be enabled
- AI assistants are not allowed
- Reexam in August
- Grade is an *overall assessment* of the implementation project and the exam
 - The result of the final exam must meet the minimum requirements for acceptance to be able to pass the course
 - The final exam contributes 80% to the final grade
- wiseflow.au.dk
 - Download .zip + add missing code + upload new .zip
- Questions? – post them on Brightspace

Content of .zip file

```
A.py - C:\Users\au121\Desktop\ipsa22\ipsa22exam\A.py (3.11.3)
File Edit Format Run Options Window Help

INTERVAL SUM

Your task is to write a function interval_sum(i, j), that returns
the sum i + (i + 1) + ... + j. Eg. for i = 10 and j = 13 the sum
returned should be 10 + 11 + 12 + 13 = 46.

Input: Two lines, containing integers i and j, respectively.
It is guaranteed that 1 <= i <= j <= 100.

Output: The sum i + (i + 1) + ... + j.

Example:

Input:  10
       13

Output: 46

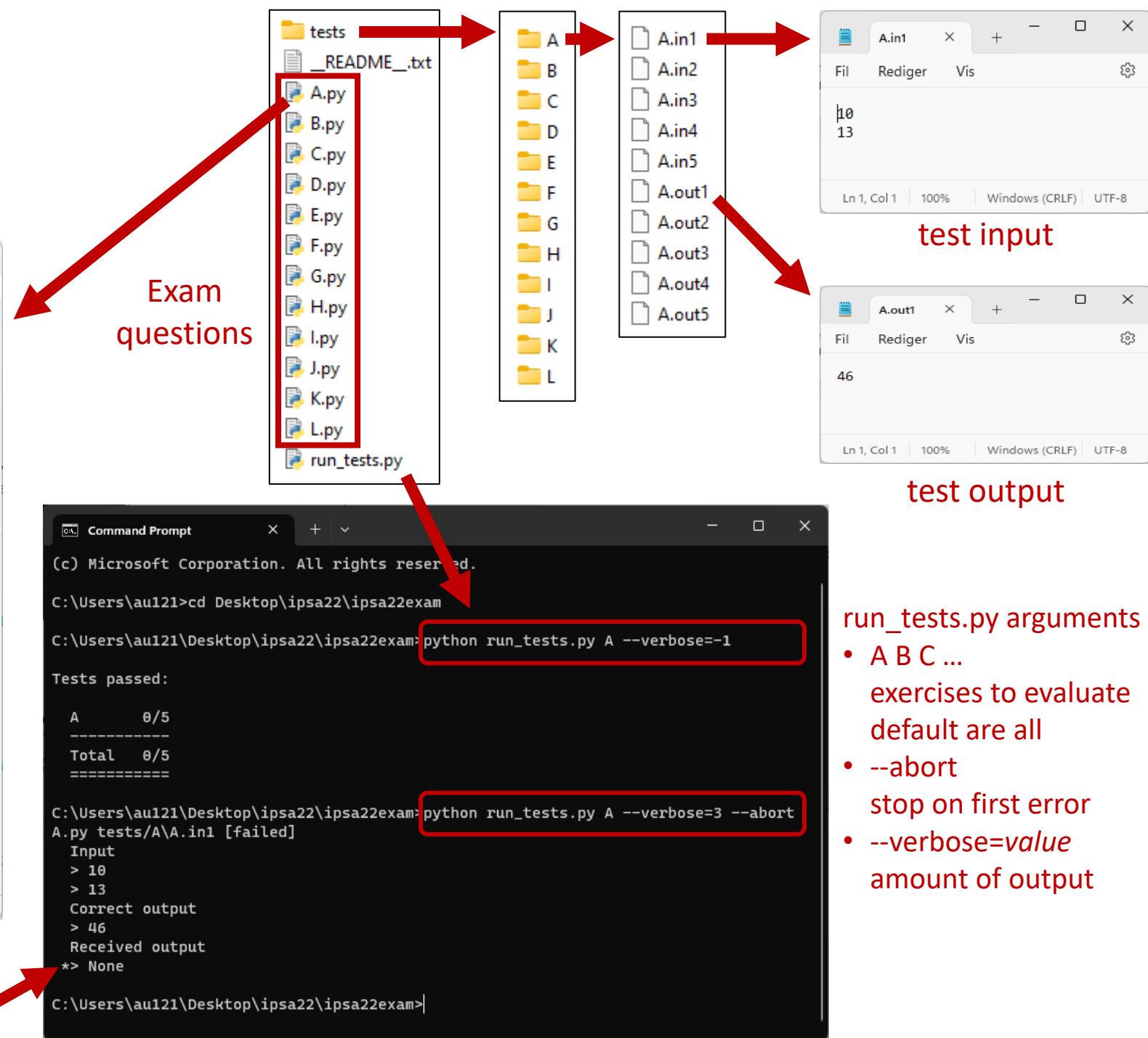
Note: The below code already handles the input and output.
...

def interval_sum(i, j):
    # insert code
    pass

i = int(input())
j = int(input())
print(interval_sum(i, j))

Ln: 1 Col: 0
```

* = line with wrong output



Using VS Code

F11 toggles full screen

Place terminal right
Right click menu > Panel position > Right

ctrl-k ctrl-o
Open folder

Click to
toggle file
explorer

Disable
AI assistants
(GitHub Copilot)



The screenshot shows the VS Code interface with the following components and annotations:

- Explorer Panel:** Shows the file structure of the 'IPSA25EXAM' folder. Files include `__README__.txt`, `A.py`, `B.py`, `C.py`, `D.py`, `E.py`, `F.py` (selected), `G.py`, `H.py`, `I.py`, `J.py`, `K.py`, `L.py`, `run_tests.log`, and `run_tests.py`. An arrow points to the Explorer icon in the sidebar with the text "ctrl-k ctrl-o Open folder". Another arrow points to the Explorer icon with the text "Click to toggle file explorer".
- Main Editor:** Displays the content of `F.py`. The code includes a task description for a `progression` function and a test case. An arrow points to the top of the editor with the text "F11 toggles full screen".
- Terminal Panel:** Positioned on the right side. It shows the output of running `python run_tests.py --verbose=-1`. The output lists test results for A.py through L.py. An arrow points to the terminal icon in the top right with the text "Place terminal right Right click menu > Panel position > Right".
- Terminal Content:**

```
PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa25exam> python run_tests.py --verbose=-1

Tests passed:

A.py    0/5
B.py    0/5
C.py    0/5
D.py    0/5
E.py    0/5
F.py    1/5
G.py    0/5
H.py    0/5
I.py    0/5
J.py    0/5
K.py    0/5
L.py    0/5

-----
Total   1/60
=====

PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa25exam> python run_tests.py --verbose=3 --abort F
F.py tests/F\F.in1 [ok]
Input
> [5, ..., 15]
Correct output
> [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
F.py tests/F\F.in2 [failed]
Input
> [4, 7, ..., 31]
Correct output
> [4, 7, 10, 13, 16, 19, 22, 25, 28, 31]
Received output
> (none)
Error
> ~~~~~
> File "C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa25exam\F.py", line 61, in progression
> return list(range(L[0], L[2] + 1)) # only works for 1st test...
> ~~~~~
> TypeError: unsupported operand type(s) for +: 'ellipsis' and 'int'
PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa25exam>
```
- Split Tab:** An arrow points to the tab bar with the text "Split tab hold 'alt' to split horizontally".
- Terminal Controls:** On the right side, text indicates "Ctrl-j Toggles terminal" and "Ctrl-l Clears terminal", with an arrow pointing to the terminal panel.

Evaluation of code

Don't expect partial scores
for this solution

```
def interval_sum(i, j):  
    if i == 10 and j == 13:  
        return 46
```

- Each problem will be assigned a **weight**
- There will be problems of **varying difficulty**
- Code will be evaluated on **known test cases** and **unknown test cases**
- In general, **automatic scoring**, in some exceptional cases manual
- Put a **comment if you copied code from exercises/slides/documentation** to avoid plagiarism

AOB ?